Mechatronics Design Laboratory I

ENME 561/661

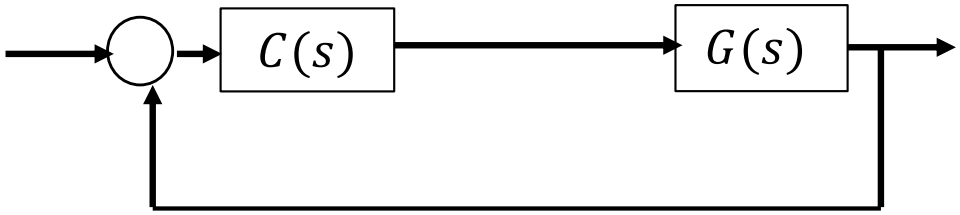
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Department of Mechanical and Manufacturing Engineering

2026/10/09

Example



Study Response Characteristics / Manual Gain Tuning using MATLAB



- Use zpk to create zero-pole-gain models

  G = zpk(zeros, poles, gain)

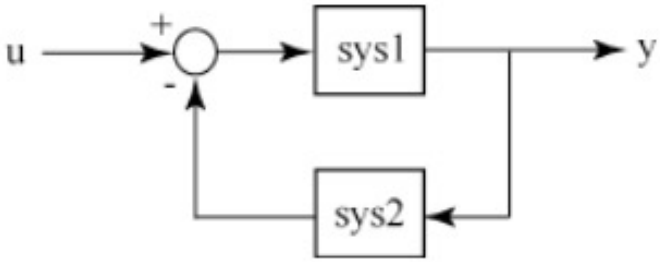
  e.g: G=zpk([],[-1 -1 -1],1);

$$: k\frac{(s-z(1))(s-z(2))\dots(s-z(m))}{(s-p(1))(s-p(2))\dots(s-p(n))}$$

- Use pid to create a parallel-form proportional-integral-derivative (PID) controller model

  C= pid(Kp, Ki, Kd, Tf)

$$C = K_p + \frac{K_i}{s} + \frac{K_d s}{T_f s + 1}$$



- Define TFs of the forward path and feedback path

  sys1 = C*G, sys2 = 1

- Use $feedback$ to return a model object sys for the negative feedback interconnection of model objects sys1,sys2

  $feedback$(sys1,sys2)

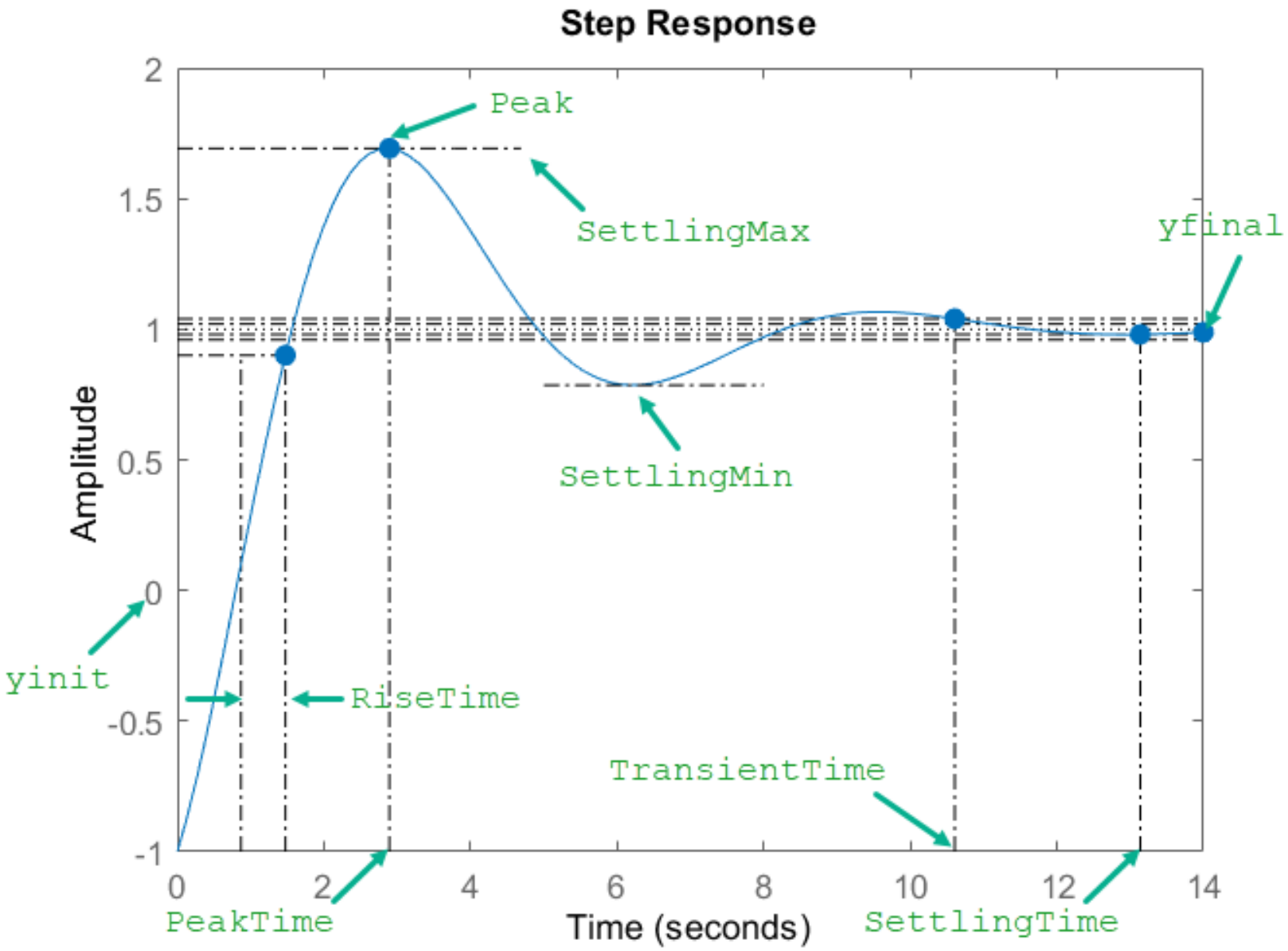
- Use $step$ to compute the step response y of the dynamic system sys

  $step$($feedback$(sys1,sys2))

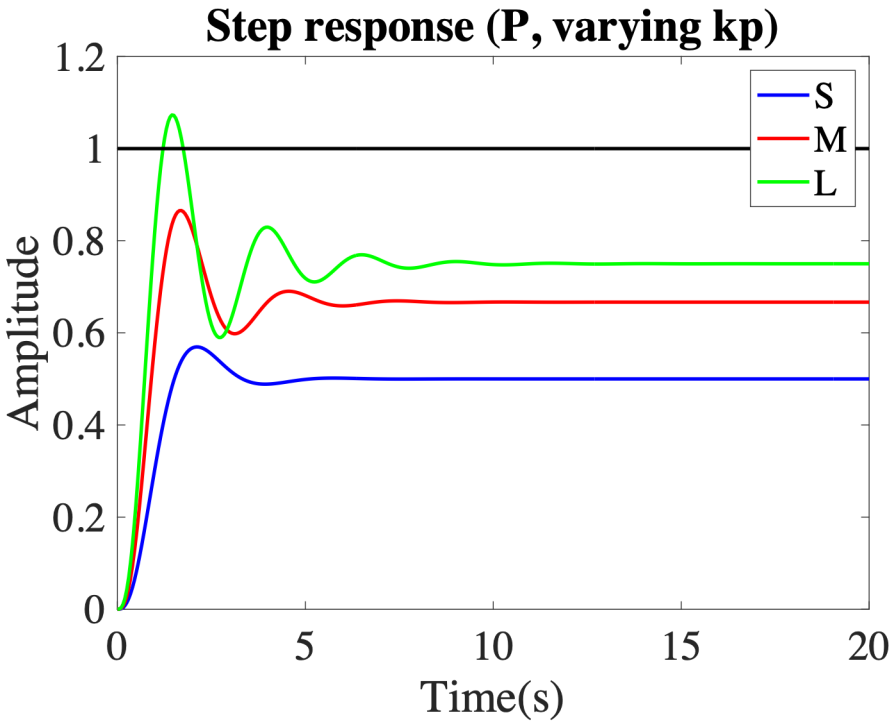
  [y,t] =$step$(feedback(sys1,sys2))

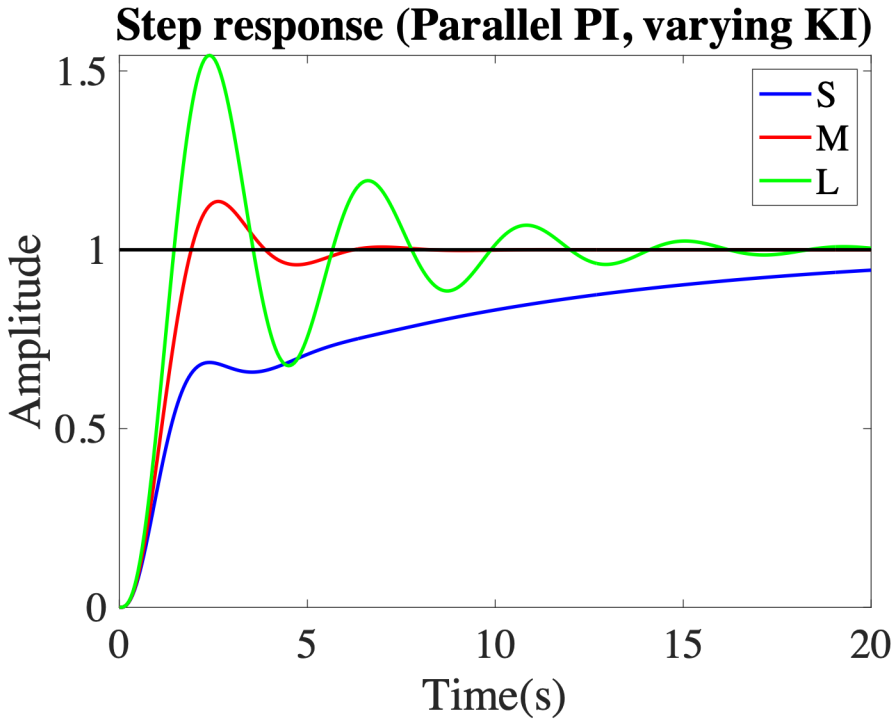
- Use $stepinfo$ to compute step-response characteristics for a dynamic system model

MATLAB stepinfo function:
Rise time, settling time, and other step-response characteristics

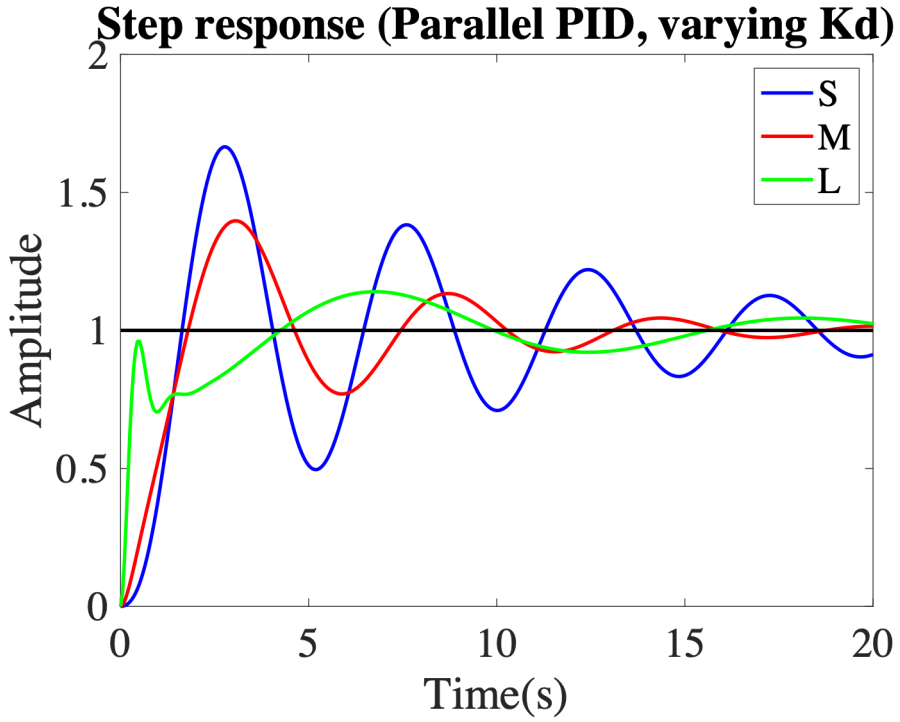


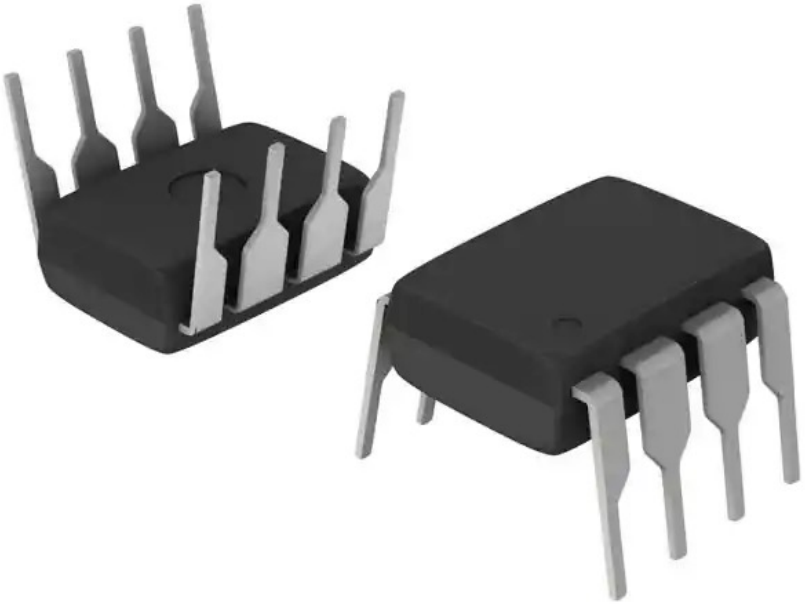
Source: https://www.mathworks.com

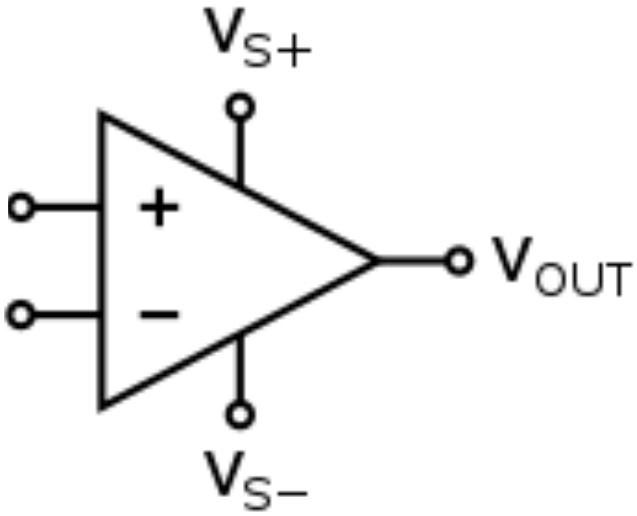




Plant TF: $G = \frac{1}{(s+1)^3}$







Operational Amplifier (op amp)

- Building blocks of modern electronic instrumentation
- Active circuit element to perform mathematical operations

Photo: https://www.digikey.ca/

Photo: https://en.wikipedia.org/

Pinout of a typical 8-pin op amp (741)

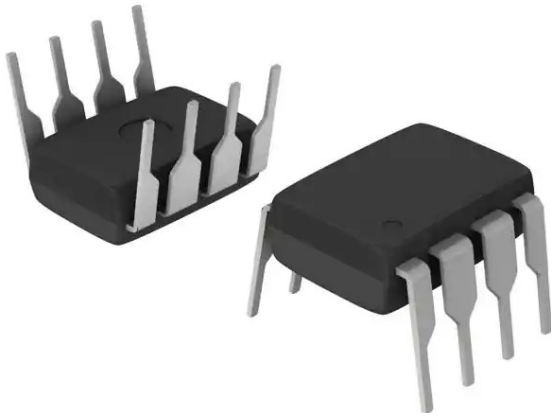
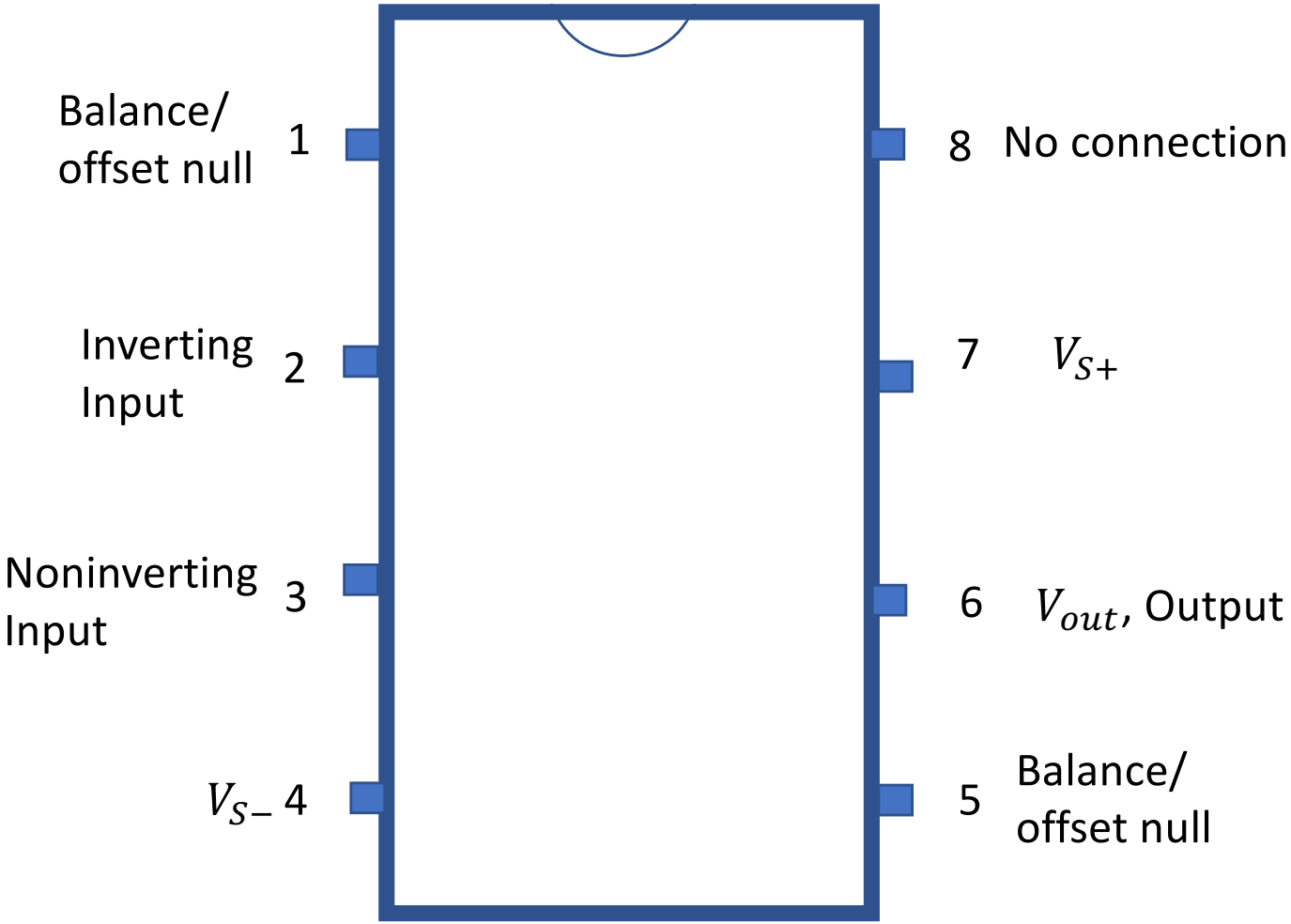
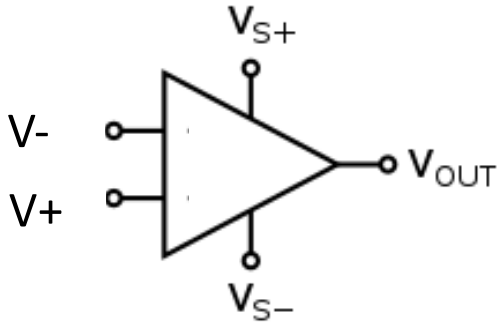


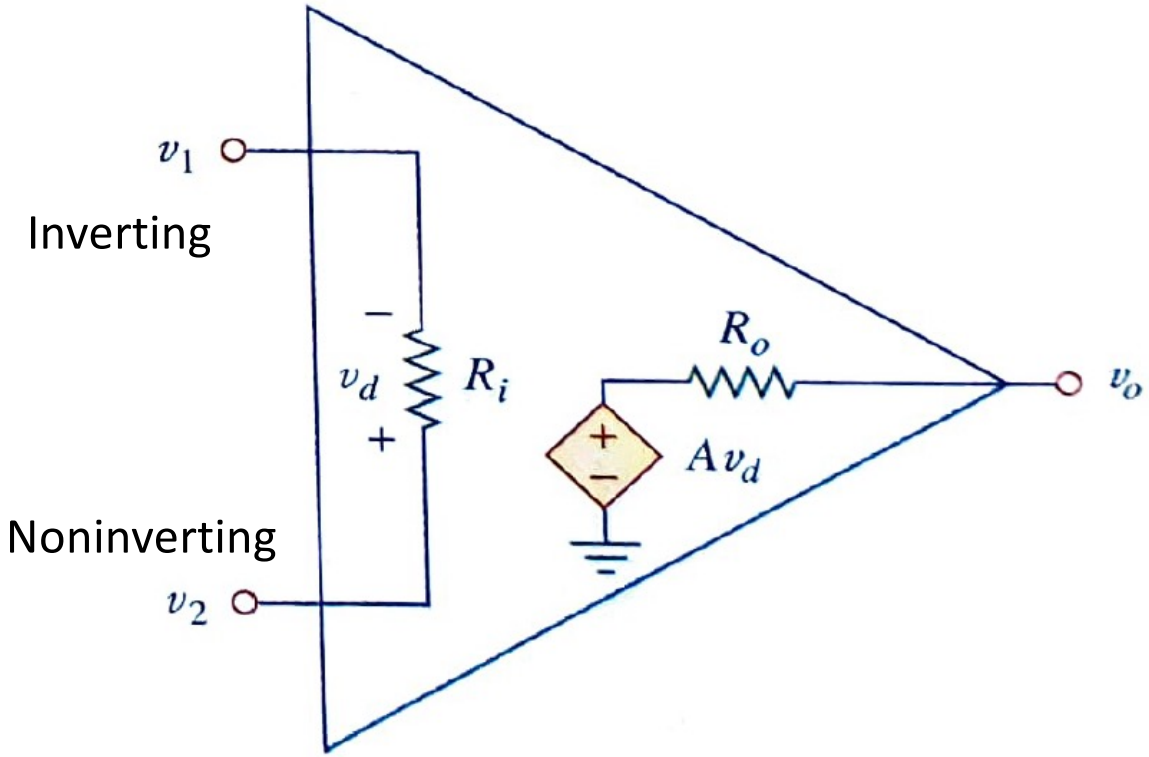
Photo: https://www.digikey.ca/

Top view





Equivalent circuit of an op amp

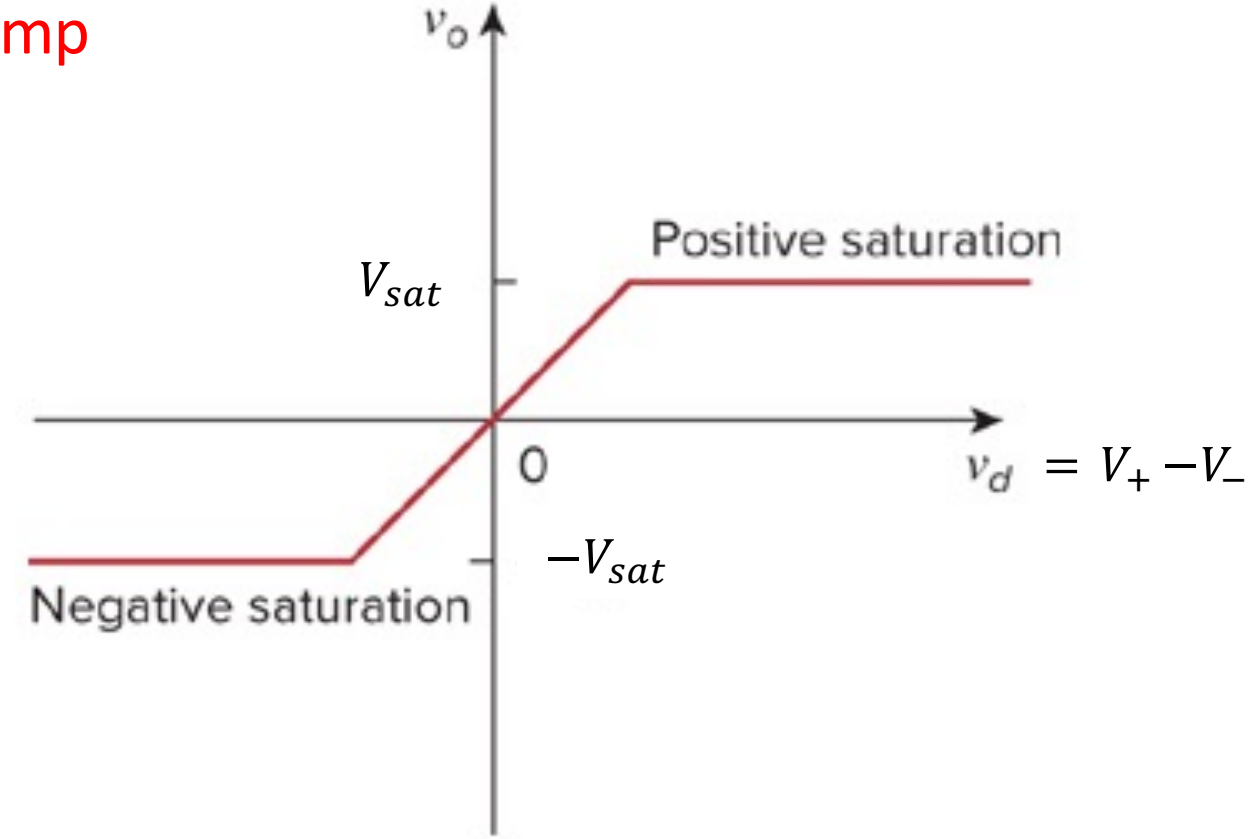


Typical values of op amp parameters

Parameter	typical range	Ideal value
Open loop gain (A)	10^5 to 10^8	∞
Input resistance (R_i)	10^5 to 10^{13} Ω	∞ Ω
Output resistance (R_o)	10 to 100 Ω	0 Ω
Supply voltage (V_{cc})	5 to 24 V	

Open-loop op amp

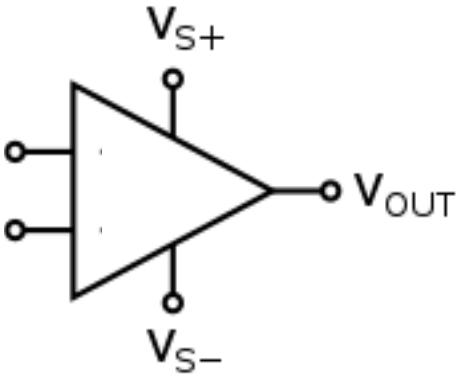
Saturation



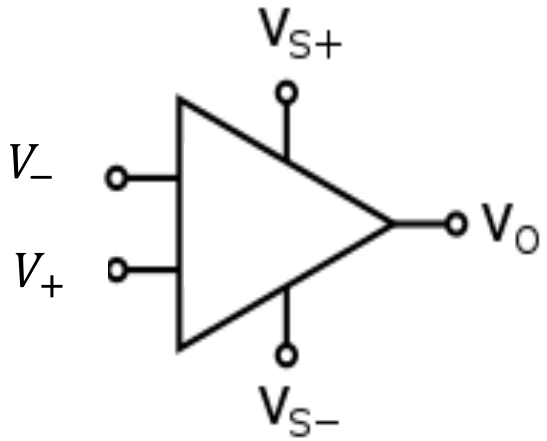
Application: Comparator op amp

$V_{ref} = V_-$

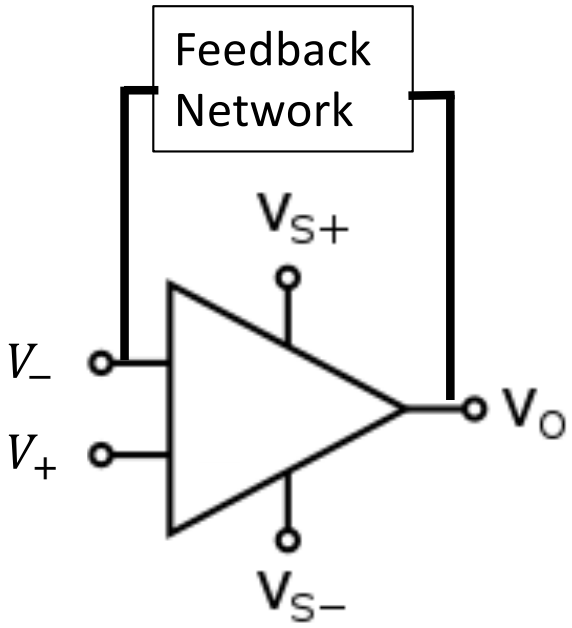
$\boldsymbol{V_i} = V_+$



- Open loop



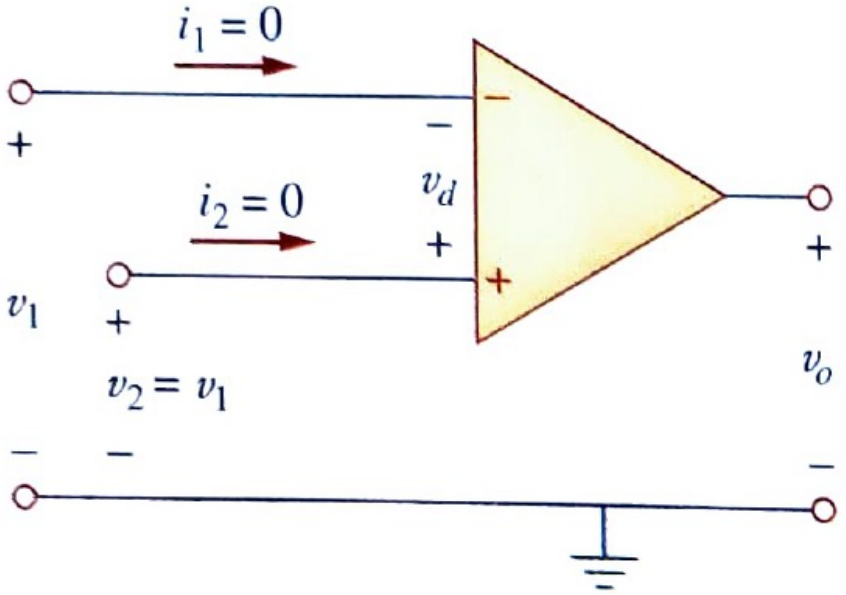
- Negative feedback



Open loop voltage gain: A

Closed loop voltage gain: $\frac{V_0}{V_{in}}$

Ideal op amp with a negative feedback loop



1. Infinite open-loop gain

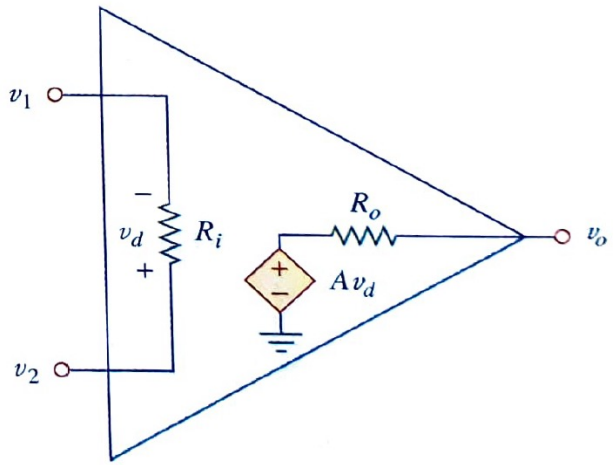
   Thus, the voltage across the input terminals is equal to zero

2. Infinite input resistance

   Thus, the currents into both input terminals are zero

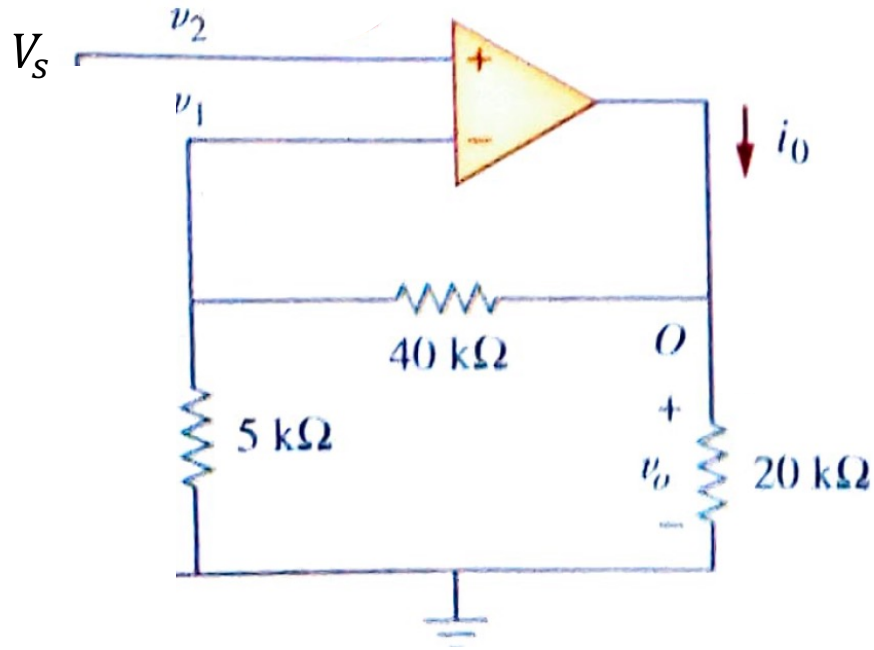
3. Zero output resistance

   Thus, the output voltage does not depend on the output current



Example 2- Ideal Op Amp

- Assume ideal op amp
- Find closed loop gain, and
- Find i_o, if v_s=1 V



Reference

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